

Task1.1

A

1. Superkeys: (EmpId), (SSN), (Email), (Phone), (EmpId, Name), (EmpId, SSN, Email)

2. Candidate keys: (EmpId), (SSN), (Email), (Phone)

3. Primary Key: (EmpId), because: SSN-is sensitive, Email and Phone can be changed so EmpId is ideal because it does not show sensitive data and do not need ability to be changed to represent employee.

4. Based only on the sample data shown, no. All three records show distinct, unique phone numbers(555-0101, 555-0102, 555-0103) they differ on sample by one digit.

B

1. (StudentID, CourseCode, Section, Semester, Year)

2.

StudentID- Required because multiple students enroll in the same courses.

CourseCode - Required because a single student can register for multiple different courses

Section – Required because courseCode might be identical in different sections and student can get more than one section at the same time

Semester and Time – Required because we need to distinguish when student get registration to course because he can do it can take it multiple times in different semesters and in same semesters but in different years like in Autumn 2025 and Autumn 2026

3. None exist if we consider that Semester is not numerical like 1st, 2nd..., but autumn, spring, summer semesters, or CourseCode is unique for all sections so we can remove sections from primary key.

Task 1.2

1. Student(AdvisorID - Foreign Key) – Professor (ProfID) {Many to one}
Professor(Department - Foreign Key) – Department (DeptCode) {Many to one}

Course(DepartmentCode - Foreign Key) – Department (DeptCode) {Many to one}

Department(ChairID - Foreign Key) – Professor(ProfID) {One to one}

Enrollment(StudentID- Foreign Key) – Student(StudentID) {Many to one}

Enrollment(CourseID- Foreign Key) – Course (CourseID) {Many to one}

Enrollment realize many to many connection between Student and Course

Part 4

Task 4.1

1. StudentID -> StudentName, StudentMajor;
ProjectID -> ProjectTitle, ProjectType, StartDate, EndDate;
SupervisorID -> SupervisorName, SupervisorDept;
StudentID, ProjectID -> Role, HoursWorked;
2. Update of student always requires to find and fill name and major which will have problems if student change their name or major we need to update all rows except from updating just one in another table same with project and supervisor.
Insert we create new student or project only when we assign them to it so it is a problem when we want to create project before and after add people to it and we cannot put NULL because Project an Student are primary keys
Delete we can erase student or project from existence if we try to delete row where person was assigned to project and we delete sensitive info about student and project too

3. No violations

4. Primary key: (StudentID, ProjectID), Partial Dependencies is:
StudentID -> StudentName, StudentMajor;
ProjectID -> ProjectTitle, ProjectType, StartDate, EndDate, SupervisorID, SupervisorName, SupervisorDept;
Decomposition:

Student(StudentID, StudentName, StudentMajor)
Project(ProjectID, ProjectTitle, ProjectType, StartDate, EndDate, SupervisorID, SupervisorName, SupervisorDept)
StudentProjectAssignment(StudentID, ProjectID, Role, HoursWorked)

5. Decomposition:
Student(StudentID, StudentName, StudentMajor)
Supervisor(SupervisorID, SupervisorName, SupervisorDept)
Project(ProjectID, ProjectTitle, ProjectType, StartDate, EndDate, SupervisorID)
StudentProjectAssignment(StudentID, ProjectID, Role, HoursWorked)

Task 4.2

1. Primary key: (StudentID, TimeSlot)
2. StudentID → StudentMajor

$\text{CourseID} \rightarrow \text{CourseName}$
 $\text{InstructorID} \rightarrow \text{InstructorName}$
 $\text{Room} \rightarrow \text{Building}$
 $\text{CourseID}, \text{TimeSlot} \rightarrow \text{InstructorID}, \text{Room}$
 $\text{StudentID}, \text{TimeSlot} \rightarrow \text{CourseID}$

3. No it is not. For BCNF, every determinant (left side of FD) must be a superkey. FDs like $\text{StudentID} \rightarrow \text{StudentMajor}$ or $\text{CourseID} \rightarrow \text{CourseName}$ violate BCNF because StudentID and CourseID are not superkeys on their own.

4. Decomposition:

1NF: No violations;

2NF:

Student(StudentID, StudentMajor)

StudentCourseSchedule(StudentID, TimeSlot, CourseID,

CourseName,

InstructorID, InstructorName, Room, Building)

3NF:

Student(StudentID, StudentMajor)

Instructor(InstructorID, InstructorName)

Course(CourseID, CourseName)

Classroom(Room, Building)

StudentCourseSchedule (StudentID, TimeSlot, CourseID,

InstructorID, Room)

BCNF:

Student(StudentID, StudentMajor)

Instructor(InstructorID, InstructorName)

Course(CourseID, CourseName)

Classroom(Room, Building)

SectionSchedule(CourseID, TimeSlot, InstructorID, Room)

StudentCourseSchedule (StudentID, TimeSlot, CourseID)

5. Potential Loss of Information used that structure loss constraint about each course section is taught by one instructor at one time in one room. Because SectionSchedule can state that instructor can do two courses at same time

Task 5.1

2. STUDENT(student_id PK, name, university_email UNIQUE)
 FACULTY(faculty_id PK, name, email UNIQUE)
 CLUB(club_id PK, club_name, description, advisor_id FK -> FACULTY.faculty_id)

CLUB_MEMBERSHIP(club_id PK/FK → CLUB.club_id, student_id PK/FK → STUDENT.student_id, joined_date, membership_status)

OFFICER_POSITION(position_id PK, title)

CLUB_OFFICER(club_id PK/FK → CLUB.club_id, student_id PK/FK → STUDENT.student_id, position_id PK/FK →

OFFICER_POSITION.position_id, term_start PK, term_end)

CLUB_EVENT(event_id PK, club_id FK → CLUB.club_id, event_name, start_datetime, end_datetime)

EVENT_ATTENDANCE(event_id PK/FK → CLUB_EVENT.event_id, student_id PK/FK → STUDENT.student_id, attendance_status, check_in_time)

ROOM(room_id PK, building, room_number, capacity)

ROOM_RESERVATION(reservation_id PK, event_id FK → CLUB_EVENT.event_id, room_id FK → ROOM.room_id, reserved_start, reserved_end)

CLUB_BUDGET(budget_id PK, club_id FK → CLUB.club_id, fiscal_year, allocated_amount, UNIQUE(club_id, fiscal_year))

EXPENSE(expense_id PK, budget_id FK → CLUB_BUDGET.budget_id, expense_date, amount, description)

3. To do officer positions can do through attributes, but chose through different table to give multiple officer positions to one person
4. Find all students who are officers in the Computer Science Club, including their position and term dates.

List all club events scheduled for next week, with their reserved room, building, and capacity.

Show each club's budget for the current fiscal year, total recorded expenses, and remaining balance.